

Write your questions
and thoughts here!



Direct Substitution		Factor and Cancel	
1. $\lim_{x \rightarrow -1} (x^2 + 2x - 4)$	2. $\lim_{x \rightarrow 2} 6$	3. $\lim_{x \rightarrow 0} \frac{4x^2 - 5x}{x}$	4. $\lim_{x \rightarrow -7} \frac{2x^2 + 13x - 7}{x + 7}$

Limit Does Not Exist

5. $\lim_{x \rightarrow -6} \frac{x^2 + 4x + 3}{x + 6}$

Special Trig Limits:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \quad \text{or} \quad \lim_{x \rightarrow 0} \frac{x}{\sin x} =$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = \quad \text{or} \quad \lim_{x \rightarrow 0} \frac{\cos x - 1}{x} =$$

6. $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$

7. $\lim_{x \rightarrow 0} \frac{\sin 7x}{\sin 9x}$

8. $\lim_{x \rightarrow 0} \frac{\cos^2 x - 1}{x(\cos x + 1)}$

1.6 Algebraic Manipulation of Limits

Calculus

Practice

Evaluate each limit.

1. $\lim_{x \rightarrow 2} (x - x^2)$	2. $\lim_{x \rightarrow 5} (x + 1)^2$	3. $\lim_{x \rightarrow 1} \frac{x^2 - 5x}{x - 1}$	4. $\lim_{x \rightarrow 1} \frac{x^2 + x - 30}{x - 1}$
5. $\lim_{x \rightarrow 0} \frac{3x}{\sin x}$	6. $\lim_{x \rightarrow 0} \frac{\sin(2x)}{3x}$	7. $\lim_{x \rightarrow -2} (3x^2 - x + 1)$	8. $\lim_{x \rightarrow 3} (2x^2 + 5x - 6)$
9. $\lim_{x \rightarrow -7} \frac{2x^3 + 11x^2 - 21x}{x^2 + 7x}$	10. $\lim_{x \rightarrow 8} \frac{x^2 + 2x - 80}{x - 8}$	11. $\lim_{x \rightarrow \frac{1}{3}} \frac{6x^2 + 13x - 5}{3x - 1}$	12. $\lim_{x \rightarrow 0} \frac{7x^2 + x}{x}$
13. $\lim_{x \rightarrow -3} 14$	14. $\lim_{x \rightarrow 0} \frac{x^2 + 2x - 8}{x - 4}$	15. $\lim_{x \rightarrow -2} \frac{x^2 - 4x - 10}{x}$	16. $\lim_{x \rightarrow 0} \frac{3x^2 + 5x}{x}$

$$17. \lim_{x \rightarrow 4} \frac{5x^2 - 21x + 4}{x - 4}$$

$$18. \lim_{x \rightarrow \frac{1}{2}} \frac{1 - x - 2x^2}{2x - 1}$$

$$19. \lim_{x \rightarrow \pi} \cos x$$

$$20. \lim_{x \rightarrow \frac{\pi}{8}} \sin(4x)$$

$$21. \lim_{x \rightarrow 2} \frac{x^2 + 6x - 16}{2 - x}$$

$$22. \lim_{x \rightarrow 5} \frac{2x^2 - 17x + 35}{5 - x}$$

$$23. \lim_{x \rightarrow 0} \frac{(1 - \cos^2 x) \sin x}{x^2}$$

1.6 Algebraic Manipulation of Limits

Test Prep

24. Evaluate $\lim_{x \rightarrow 1} \frac{\ln x}{3x}$ is

- (A) 0 (B) $\frac{3}{e}$ (C) e (D) 3 (E) The limit does not exist.

25. $\lim_{x \rightarrow 0} 4 \frac{\sin x \cos x - \sin x}{x^2}$ is

- (A) 2 (B) $\frac{40}{3}$ (C) ∞ (D) 0 (E) undefined

26. $\lim_{x \rightarrow a} \frac{x^2 - 2ax + a^2}{x - a} =$

- (A) $-\infty$ (B) a (C) 0 (D) ∞ (E) The limit does not exist.

27. $\lim_{x \rightarrow 0} \left(\frac{3x^2 + 5\cos x - 5}{2x} \right) =$

- (A) 0 (B) $\frac{5}{2}$ (C) 3 (D) 4 (E) Does not exist